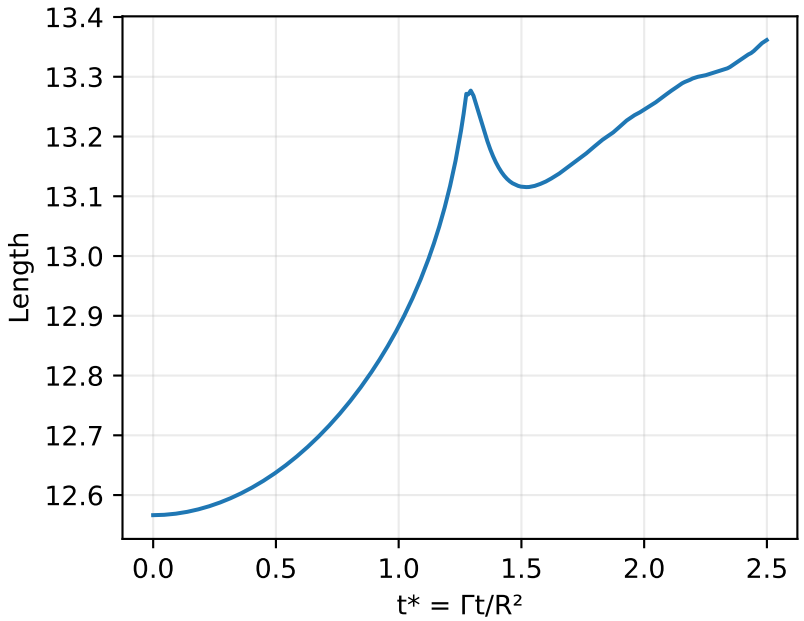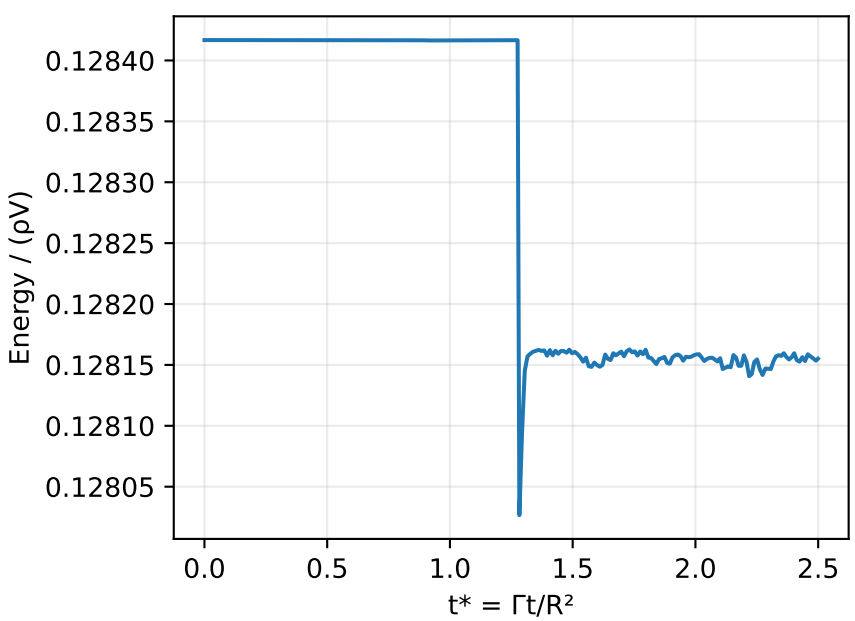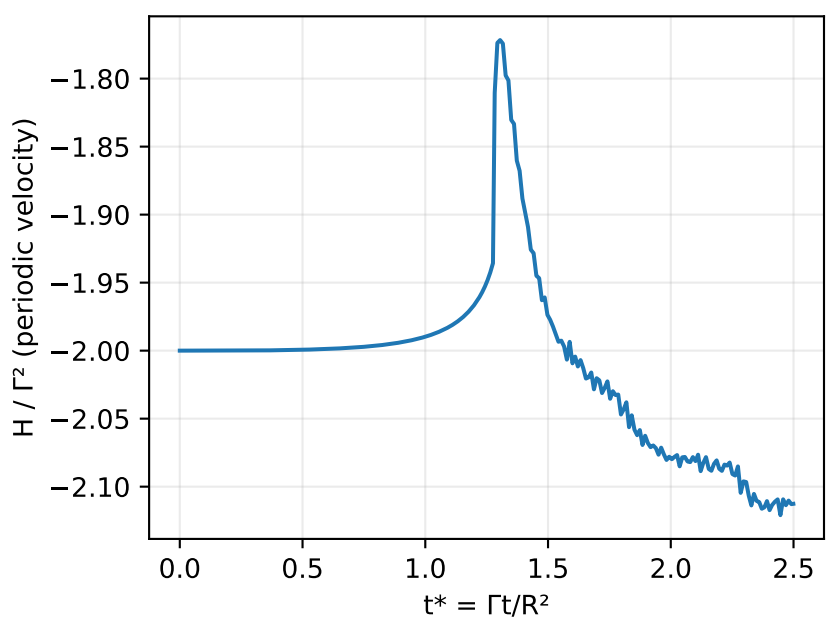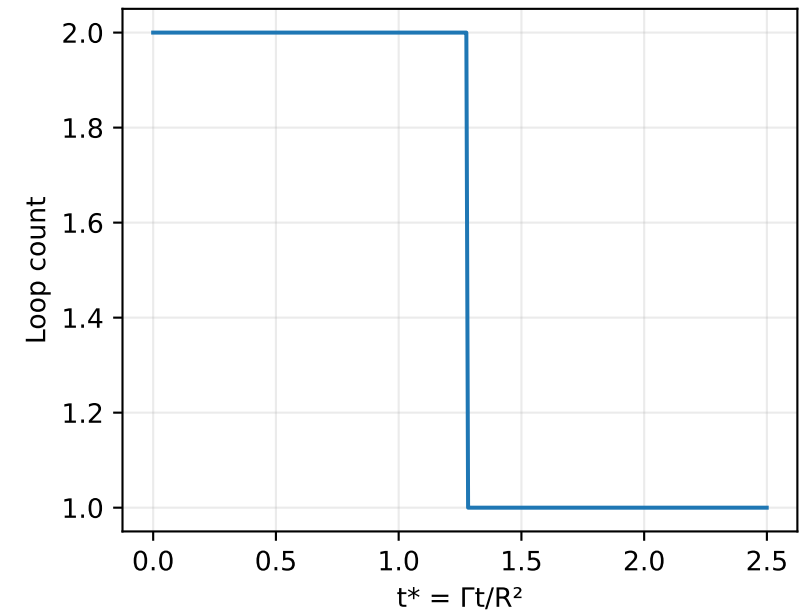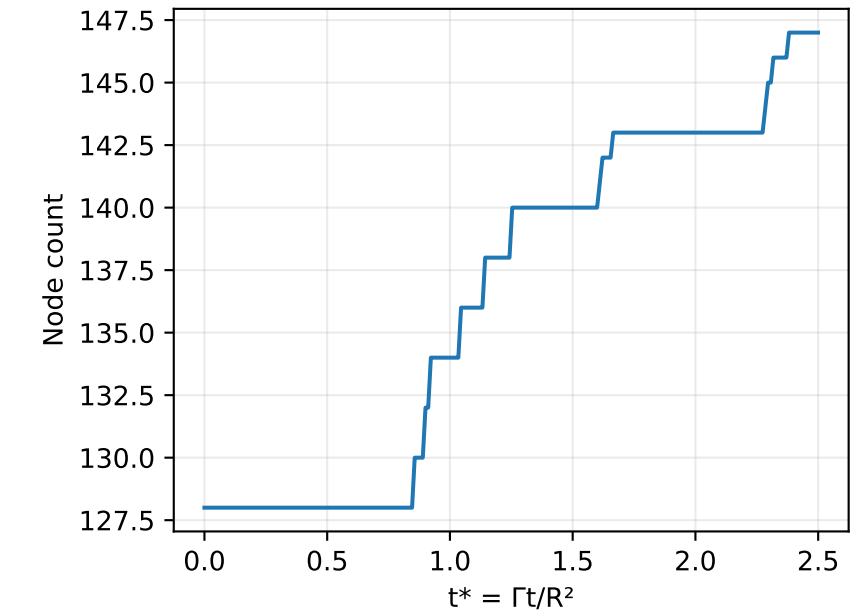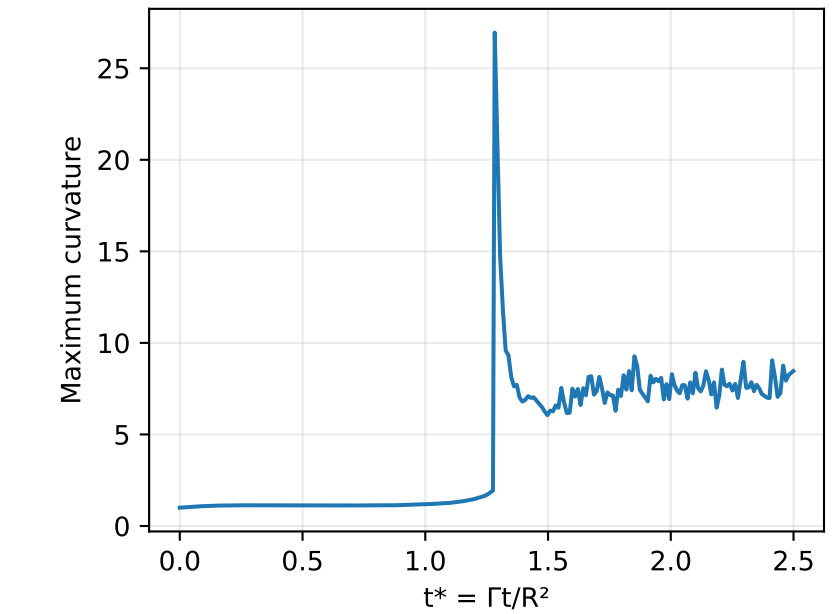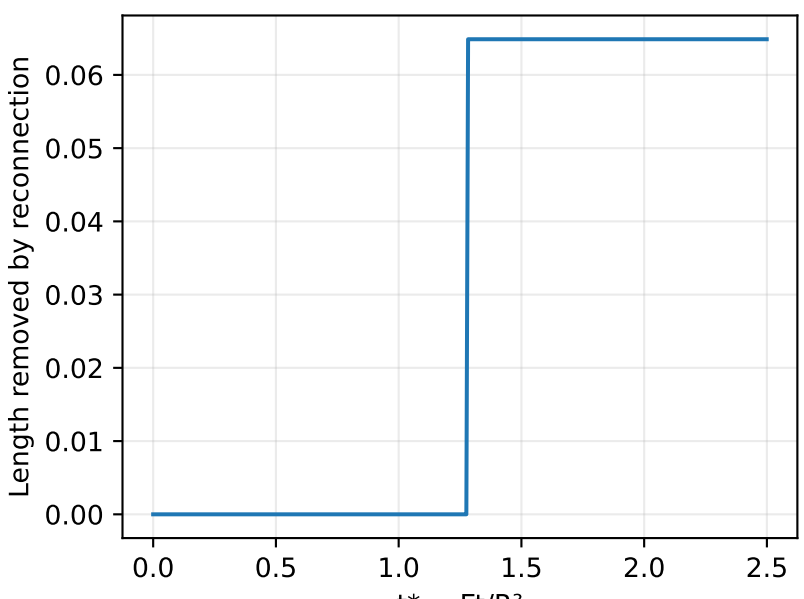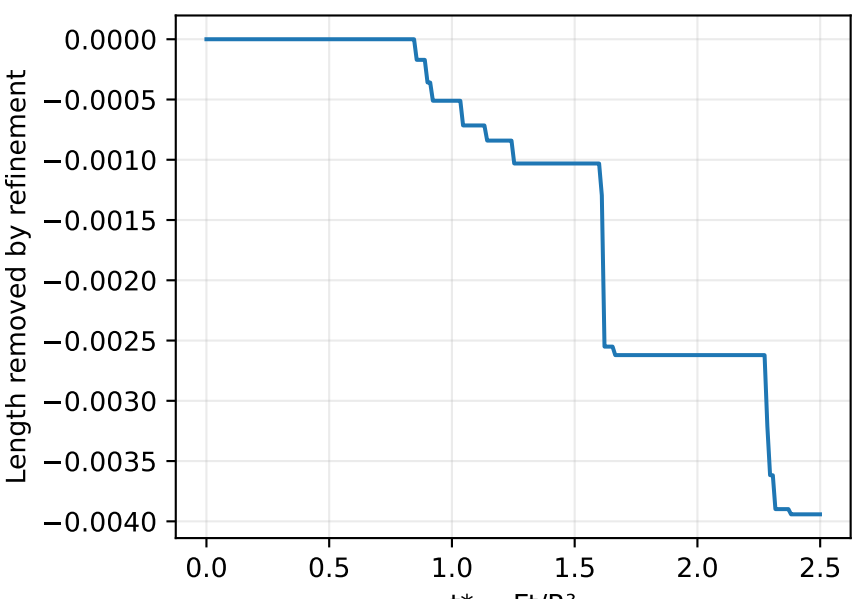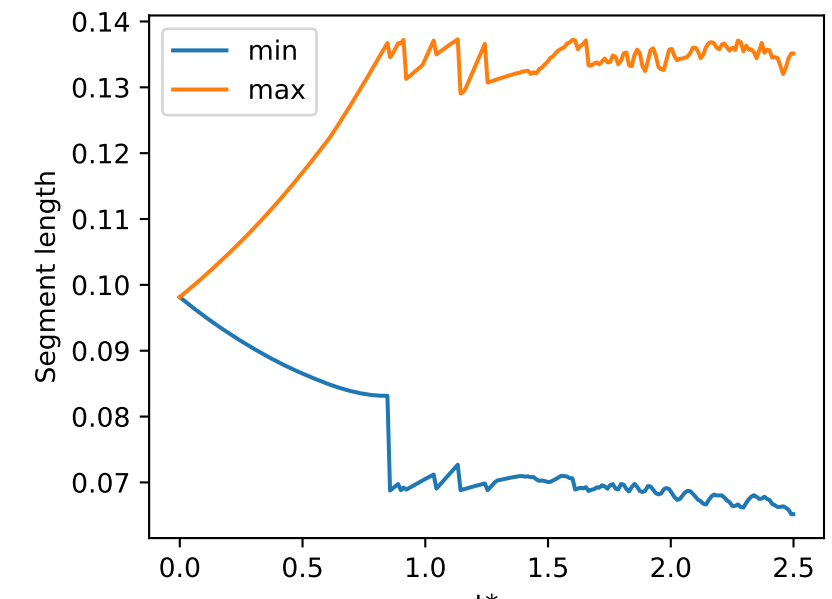hopf_base · a=0.001 · no mutual friction